



**HỌC VIỆN
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Web Design

Lecture 5 - CSS box model



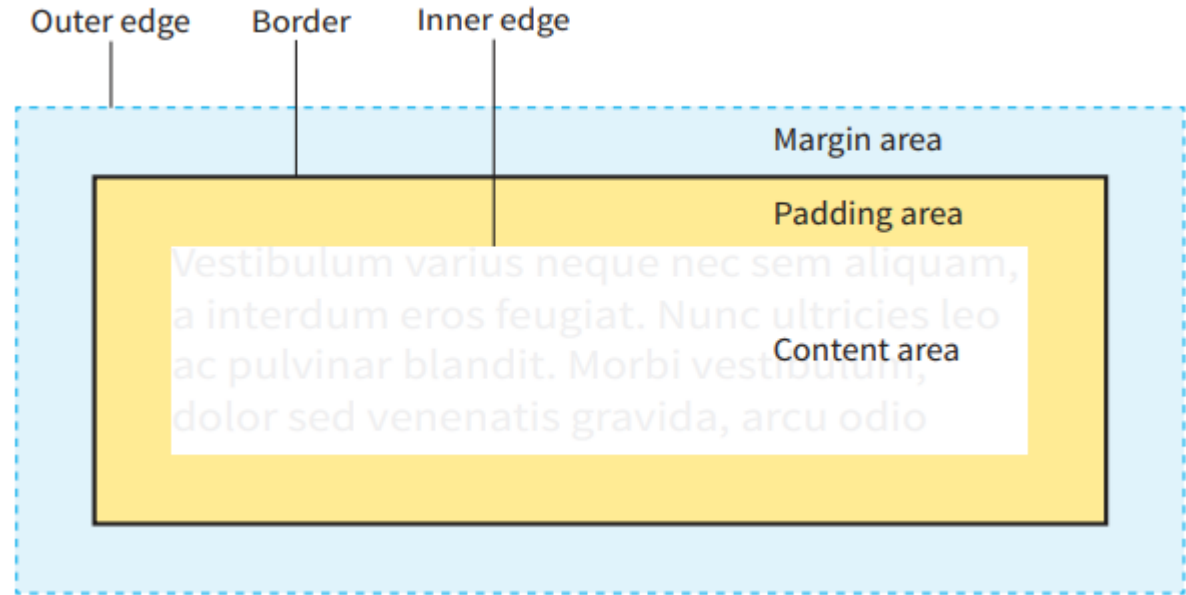
- 4.1. The box model
 - 4.2. Width and height
 - 4.3. Distance measures
 - 4.4. Margin and padding
 - 4.5. Borders
 - 4.6. Alignment
 - 4.7. Position attributes
 - 4.8. Overflow, float and display
-
-



Both block-level
and inline,
generates a
***rectangular
element box***

Hands-on

**Exercises &
Homeworks /
Lecture 5 /
box_model.html**





- Width
- Height
- Padding
- Border
- Margin

4.2. Width and height



Width and height of a box are number values in px, cm or:

- **auto**: default value, calculated by the browser
- **%**: calculated by % of the outer box
- **inherit**: calculated by the father/ancestor

Box-sizing: content-box | border-box

```
<html lang="en">
<head>
  <meta charset="UTF-8">
  <style>
    div {
      height: 100px;
      width: 50%;
      background-color: #1abc9c;
      text-align: center;
      color: white;
      padding-top: 30px;
    }
  </style>
</head>
<body>
  <div>
    <div>
      <h1>Tiêu đề h1</h1>
      <h2>Tiêu đề h2</h2>
      <p>Đoạn văn 1</p>
      <p>Đoạn văn 2</p>
    </div>
  </div>
</body>
</html>
```



Sizing the content-box: Textbook, page 357

Sizing the border-box: Textbook, page 358

Exercises & Homeworks / Lecture 5 / sizing.html

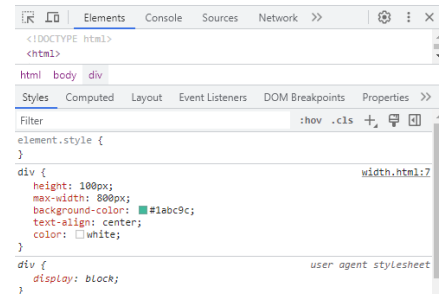
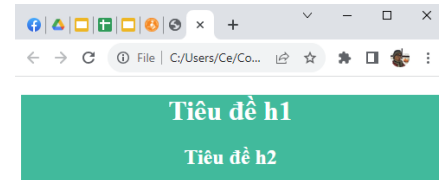


The maximum width and height values

Normal **width**: a scroll bar will appear in the browser if the screen is too small

Max-width: the box will be resized if the screen is not large enough

```
div {  
  height: 100px;  
  max-width: 800px;  
  background-color: #1abc9c;  
}
```





Exercises & Homeworks / Lecture 5 / max_width.html

4.3. Distance measures



- Absolute distance
- Relative distance



- Absolute distance
 - **cm: centimeters**
 - mm: millimeters
 - in: inches (1in = 96px = 2.54cm)
 - **px: pixels (1px = 1/96th of 1in)**
 - **pt: points (1pt = 1/72 of 1in)**
 - pc: picas (1pc = 12 pt)

Further reading on the reference book: ***Responsive web design with HTML5 and CSS***



- Relative distance:
 - **em: relative with the current font size**
 - rem: Relative to font-size of the root element
 - ex: relative with the x-height of the current font size.
 - **%: relative with the parent element**
 - ch, vw, vh, vmin, vmax

Relative length units scale better between different rendering medium.



Exercises & Homeworks / Lecture 5 / relative_distance.html

```
<style>
p {
  font-size: 16px;
  line-height: 2em;
}
```

These paragraphs have a calculated line-height of: $2 \times 16\text{px} = 32\text{px}$.

These paragraphs have a calculated line-height of: $2 \times 16\text{px} = 32\text{px}$.

```
div {
  font-size: 30px;
  border: 1px solid black;
}
```

These paragraphs have a calculated line-height of: $2 \times 16\text{px} = 32\text{px}$.

```
span {
  font-size: 0.5em;
}
</style>
```

The font-size of the div element is set to 30px. The span element inside the div element has a font-size of 0.5em, which equals to $0.5 \times 30 = 15\text{px}$.



Margins are used to create **space around** elements, outside of any defined borders.

CSS has properties for specifying the margin for each side of an element:

margin-top, margin-right, margin-bottom, margin-left



All the margin properties can have the following values:

- **auto** - the browser calculates the margin
- **length** - specifies a margin in px, pt, cm, etc.
- **%** - specifies a margin in % of the width of the containing element
- **inherit** - specifies that the margin should be inherited from the parent element

Notice: **Negative values** are allowed.

Exercises & Homeworks / Lecture 5 / margin.html



If the margin property has four values:

margin: 25px 50px 75px 100px;

- top margin is 25px
- right margin is 50px
- bottom margin is 75px
- left margin is 100px

Question: How about a margin with 3, 2, and even only 1 value?

E.g.: *margin: 25px 50px;*



Padding is used to create space around an element's content, **inside** of any defined borders.

Hands-on:

Exercises & Homeworks / Lecture 5 / padding.html



The CSS width property specifies the width of the element's content area. The content area is the ***portion inside*** the padding, border, and margin of an element.

So, if an element has a specified width, the padding added to that element will be ***added to the total width*** of the element.

https://www.w3schools.com/css/tryit.asp?filename=trycss_padding_width

This div is 300px wide.

The width of this div is 350px, even though it is defined as 300px in the CSS.



A border is simply a line drawn around the content area and its (optional) padding.

Border Style

- Values: ***none*** / ***solid*** / ***hidden*** / ***dotted*** / ***dashed*** / ***double*** / ***groove*** / ***ridge*** / ***inset*** / ***outset***
- Default: ***none***

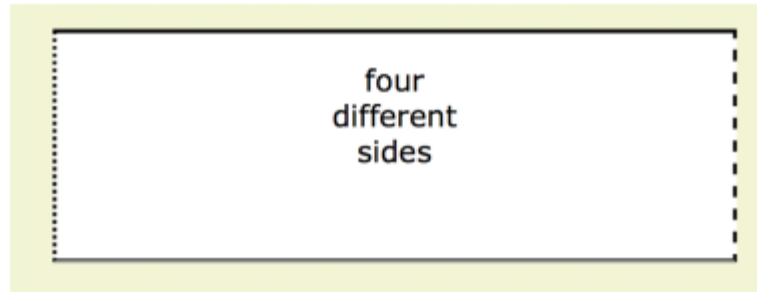
Hands-on:

https://www.w3schools.com/css/tryit.asp?filename=trycss_border-style



The border-style shorthand property works on the clockwise (TRouBLe) system described for padding earlier.

border-style: solid dashed double dotted;

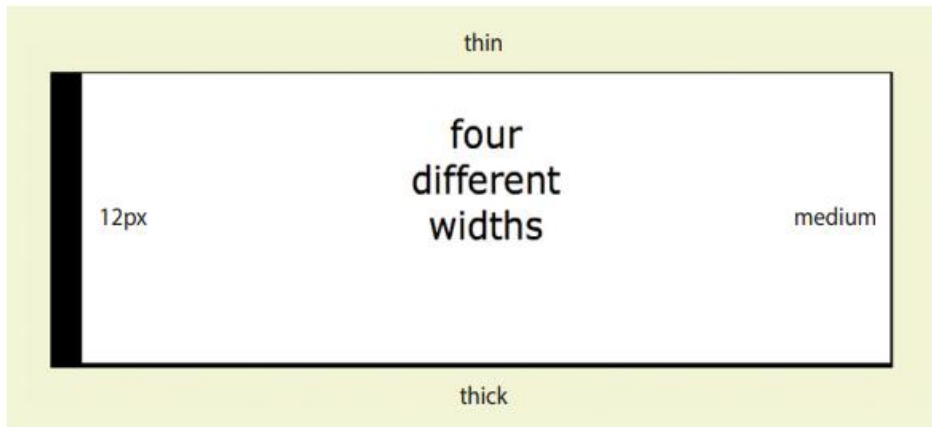




border-width

- Values: length | thin | medium | thick
- Default: medium

```
div#help {  
  border-top-width: thin;  
  border-right-width: medium;  
  border-bottom-width: thick;  
  border-left-width: 12px;  
  border-style: solid;  
  width: 300px;  
  height: 100px;  
}
```

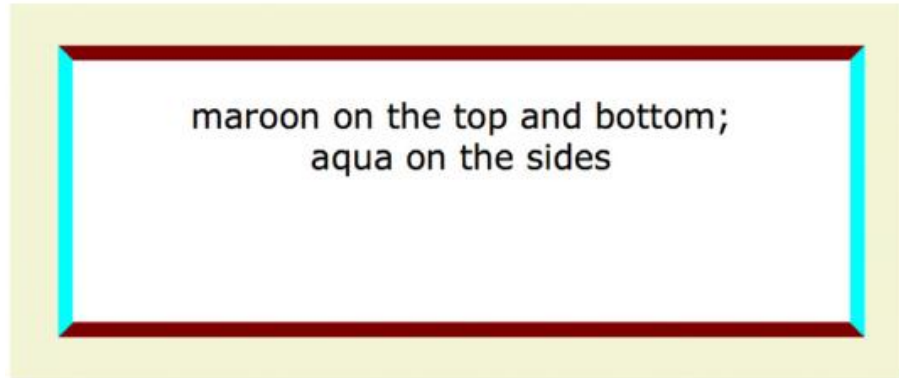




border-color

- Values: color name or RGB/HSL value | transparent
- Default: the value of the color property for the element

```
div#special {  
  border-color: maroon aqua;  
  border-style: solid;  
  border-width: 6px;  
  width: 300px;  
  height: 100px;  
}
```





Further reading, page 371

```
p {  
  width: 200px;  
  height: 100px;  
  background: darkorange;  
}
```



```
border-radius: 1em;
```



```
border-radius: 50px;
```



Page 364, 373 and 381

Further reading:

- Border image, page 375
- Box drop shadow, page 382



Test yourself, page 384



The align attribute is used to enhance the position of the element or the content inside an element. In the example below, all the elements are not aligned, **by default, they will be put on the left.**

```
<div>  
  <h1>Tiêu đề h1</h1>  
  <h2>Tiêu đề h2</h2>  
  <p>Đoạn văn 1</p>  
  <p>Đoạn văn 2</p>  
</div>
```



To horizontally center a block element (like `<div>`)
use ***margin: auto;***

Setting the ***width*** of the element will prevent it from stretching out to the edges of its container.

The element will then take up the specified width, and the remaining space will be split equally between the two margins

Hands-on:

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_container



To just center the text inside an element, use ***text-align: center;***

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_text



To center an image, set left and right margin to auto and make it into a block element:

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_image



One method for aligning elements is to use ***position: absolute;***

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_pos

Another method for aligning elements is to use the ***float*** property:

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_float



There are many ways to center an element vertically in CSS. A simple solution is to use **top** and **bottom padding**:

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_padding

To center both vertically and horizontally, use padding and **text-align: center**:

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_padding2

Another trick is to use the **line-height** property with a value that is equal to the **height** property:

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_line-height



If padding and ***line-height*** are not options, another solution is to use positioning and the ***transform*** property:

https://www.w3schools.com/css/tryit.asp?filename=trycss_align_transform

You can also use ***flexbox*** to center things. Just note that flexbox is *not supported in IE10 and earlier versions*:

https://www.w3schools.com/css/tryit.asp?filename=trycss3_align_flex



The ***position*** property specifies the type of positioning method used for an element (*static, relative, fixed, absolute or sticky*).

Elements are then positioned using the ***top, bottom, left, and right*** properties. However, these properties will not work unless the position property is set first. They also work differently depending on the position value.



HTML elements are positioned ***static*** by default.

Static positioned elements are not affected by the ***top, bottom, left, and right*** properties.

An element with ***position: static;*** is not positioned in any special way; it is always positioned according to the normal flow of the page:

https://www.w3schools.com/css/tryit.asp?filename=trycss_position_static



An element with ***position: relative;*** is positioned relative to its normal position.

Setting the ***top, right, bottom, and left*** properties of a relatively-positioned element will cause it to be adjusted away from its normal position. Other content will not be adjusted to fit into any gap left by the element.

https://www.w3schools.com/css/tryit.asp?filename=trycss_position_relative



An element with ***position: fixed;*** is positioned relative to the viewport, which means it always stays in the same place even if the page is scrolled. The top, right, bottom, and left properties are used to position the element.

A fixed element does not leave a gap in the page where it would normally have been located.

https://www.w3schools.com/css/tryit.asp?filename=trycss_position_fixed



An element with ***position: absolute;*** is positioned relative to the nearest positioned ancestor (instead of positioned relative to the viewport, like fixed).

However; if an absolute positioned element has no positioned ancestors, it uses the document body, and moves along with page scrolling.

Note: Absolute positioned elements are removed from the normal flow, and can overlap elements.

https://www.w3schools.com/css/tryit.asp?filename=trycss_position_absolute



https://www.w3schools.com/css/tryit.asp?filename=trycss_image_text_top_left

Add some text to an image in the top left corner:





The CSS **overflow** property controls what happens to content that is too big to fit into an area.

The overflow property has the following values:

- *visible* - Default. The overflow is not clipped. The content renders outside the element's box
- *hidden* - The overflow is clipped, and the rest of the content will be invisible
- *scroll* - The overflow is clipped, and a scrollbar is added to see the rest of the content
- *auto* - Similar to scroll, but it adds scrollbars only when necessary

https://www.w3schools.com/css/tryit.asp?filename=trycss_overflow_intro



The CSS **float** property specifies how an element should float.

The CSS clear property specifies what elements can float beside the cleared element and on which side.

The **float** property can have one of the following values:

- *left* - The element floats to the left of its container
- *right* - The element floats to the right of its container
- *none* - The element does not float (will be displayed just where it occurs in the text). This is default
- *inherit* - The element inherits the float value of its parent

https://www.w3schools.com/css/tryit.asp?filename=trycss_layout_float3

4.8. Overflow, float and display



The ***display*** property is the most important CSS property for controlling layout.

Every HTML element has a default display value depending on what type of element it is. The default display value for most elements is block or inline.

https://www.w3schools.com/css/css_display_visibility.asp



Page 414, 416



Test yourself, page 417
